The Impact of Social Media on Students' Mental Health

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Abstract

This study aims to assess the impact of social media on students' mental health, particularly in relation to anxiety and depression. A quantitative research method was used; data were collected through a structured questionnaire completed by 236 students from the Free University of Tbilisi and the Agricultural University of Georgia. Non-probability sampling was employed, and the data were analyzed using regression analysis, ANOVA, the chi-square test, and correlation analysis. Contrary to expectations, no association was established between time spent on social media and levels of anxiety or depression. However, a statistically significant association was identified between FOMO (fear of missing out on important events) and perceived levels of anxiety and depressive symptoms. Students who frequently encountered negative content also experienced emotional distress more often. With regard to temporarily abstaining from social media (a detox), the study did not confirm that restricting its use led to an actual reduction in anxiety and depression scores. Accordingly, the impact of social media on mental health is more closely related to its content and emotional aspects than to the intensity of use in terms of time or restrictions on that use.

Keywords: social media, students, mental health, anxiety, depression, FOMO, social anxiety, negative content, social media detox.

Rationale for the Study

For many people, checking Instagram, Facebook, TikTok, and other social media platforms has become part of everyday life. In 2023, approximately 4.9 billion people worldwide used social media. On average, each user spends 145 minutes per day on social networking platforms.[1]

Social media platforms are popular spaces where people share personal experiences, seek information, and support one another. However, caution is necessary, as numerous studies suggest that social media use may have a negative impact on mental health.[2]

[1] "Social Media's Impact on Our Mental Health and Tips to Use It Safely," *health,* November 27, 2024, https://health.ucdavis.edu/blog/cultivating-health/social-medias-impact-our-mental-health-and-tips-to-use-it-safely/2024/05#:~:text=Social%20media%20can%20negatively%20impact,evolve%20and%20become%20more%20invasive.

[2]John A Naslund et al., "Social Media and Mental Health: Benefits, Risks, and Opportunities for Research and Practice," *Journal of technology in behavioral science*, April 20, 2020, https://pmc.ncbi.nlm.nih.gov/articles/PMC7785056/.

Social media is considered one of the principal factors contributing to the deterioration of adolescent mental health, a concern recently reflected in the U.S. Surgeon General's public health advisory. A growing body of evidence indicates a causal effect of social media on mental health, drawing on unique longitudinal data collected before the COVID-19 pandemic and at two different stages of the pandemic.

The study shows that the effect of social media four months after the onset of the pandemic, under conditions of social distancing, was statistically insignificant. However, 18 months later, when college instruction had largely returned to normal, a significant negative effect of social media emerged. By analyzing data on physical activity, sleep, stress, and social support, we find evidence that, during the later stage of the pandemic, social media use reduced students' engagement in activities that promote better mental health. This effect was not observed during the early stage of the pandemic.[3]

Literature Review

Although social media can facilitate identity formation and the development of virtual connections despite physical distance, it can also adversely affect mental health and self-esteem through negative feedback, insufficient social validation, unhealthy competition, and comparisons with others.

A study involving Norwegians aged 16 to 88 demonstrated an association between anxiety levels and social media addiction. Similarly, another study conducted in the United States with 563 participants aged 18–22 showed that the amount of time spent on social media was directly associated with levels of dispositional anxiety. In addition, participants who devoted more time to social media each day scored above the clinical anxiety threshold, indicating potentially undetected clinical anxiety disorders.[4]

In their work, Durbano and Marchesi discuss the negative consequences of excessive social media use. According to research published by the Pew Research Center, 71% of young people use more than one social networking platform, which was associated with mental health problems in both adolescents and children. The study showed that spending more than two

[3] Fruehwirth JC;Weng AX;Perreira KM;, "The Effect of Social Media Use on Mental Health of College Students during the Pandemic," Health economics, June 14, 2024, https://pubmed.ncbi.nlm.nih.gov/38873817/.

[4] Ujala Zubair, Muhammad K Khan and Muna Albashari, "Link between excessive social media use and psychiatric disorders", *National Library of Madicine*, March 28,2023. https://pmc.ncbi.nlm.nih.gov/articles/PMC10129173/

hours on social networking platforms was associated with severe anxiety, poorer mental health, and suicidal thoughts.[5]

“Facebook depression” is a term used in an article published by the American Academy of Pediatrics to describe the effects of social media on young people’s health. According to the article, depression results from individuals’ excessive use of social media.

Excessive use of social networking platforms, including Facebook, is associated with numerous problems, including anxiety, depression, and loneliness. It is also noteworthy that dependence on these platforms resembles gambling addiction in many respects.

A total of 2.80 billion people worldwide use Facebook. Furthermore, more than 350 million people are estimated to meet the medical criteria for addiction. One study showed that Facebook addiction and drug addiction stimulate the same regions of the brain.[6]

Ethan Kross and his colleagues examined the relationship between Facebook use and dissatisfaction with life. They found that frequent Facebook use over a two-week period was associated with lower life satisfaction. Although this does not directly imply depression, it points to negative emotional states that may lead to depression.

Charlotte Blease notes that Facebook creates an excessively positive impression of users’ friends, encouraging social comparisons and increasing the risk of negative self-evaluation. Similarly, a study by Lup, Rosenthal, and Trub found a correlation between following large numbers of strangers on Instagram and symptoms of depression. According to the same study, positive social comparisons were associated with fewer depressive symptoms.[7]

The UCL/Kent study, “Safer Scrolling,” concludes that social media algorithms present harmful content in an entertaining format, further increasing its dissemination and influence. The findings indicate that toxic, hateful, and negative content is deliberately delivered to young users. Boys constitute a particularly high-risk group, as their mental health, specifically anxiety disorders, is further aggravated by negative content and algorithms.[8]

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[5] Federico Durbano, Barbara Marchesi, *New Developments in Anxiety Disorders* (London: InTechOpen,2016),131-132.
[6] Nahid Salma, Farhaduzzaman Bhuiyan,” Facebook addiction and its impact on depression: a cross sectional study”, *Journal of Health, Population and Nutrition,*2024.
https://jhpn.biomedcentral.com/articles/10.1186/s41043-024-00556-w
[7] Sean Hughes, *The Effects of Social Media on Depression Anxiety and Stress* (Dublin,2018),10-11.
[8] Sally Weale, *Social media algorithms’ amplifying misogynistic content,* (The guardian,2024)

Social media algorithms that expose vulnerable young people, particularly adolescents, to negative content are associated with an increase in mental health problems, including body image concerns, eating disorders, and suicidal thoughts, which lead to more serious problems, such as persistent anxiety and depression.[9]

Social media platforms have undergone a significant transformation in recent years, particularly through the shift from chronological news feeds to algorithmic recommendation systems. Algorithms take individual interests into account, making platforms more appealing. This personalization increases user engagement, time spent on the platform, and dependence, or so-called "addiction." A study found that, following the introduction of an algorithm on a social media platform, adolescents' mental health index deteriorated by an average of 0.394 standard deviations, an effect almost comparable to that of losing a job. The study also shows that, after the algorithm's introduction, adolescents more frequently reported that emotional difficulties interfered with their daily activities, social relationships, and academic or professional work. Of particular concern is that, during this period, boys substantially reduced the time they spent with friends, family, and neighbors, and in public places such as cafés and bars, while girls reported a deterioration in the quality of their social relationships. Drawing on data from the *Dutch Longitudinal Internet Studies for the Social Sciences (LISS)* panel, the analysis indicates that Instagram's introduction of an algorithm in 2016 significantly worsened adolescents' mental health, as reflected in increased anxiety.

In the mid-2000s, as social media became substantially more popular, a deterioration in adolescent mental health was observed. Although the precise causes of this trend have not yet been fully established, evidence indicates that social media algorithms play an important role in the deterioration of mental health among adolescents and adults.[10]

As mobile social networking has grown in popularity, FoMO (fear of missing out) has become a significant problem. Correlation analysis shows a positive association between mobile social media (MSM)-related FoMO and depressive symptoms, suggesting that FoMO may disrupt individuals' metacognitive functioning and thereby create conditions conducive to social media addiction. High FoMO also produces anxiety, increasing the risk of negative emotional states,

[9] Nancy Costello, *Algorithms, addiction and adolescent mental health,* (American Journal of Law and medicine 2024)

[10] Simona Mandile, *the dark side of social Media: Recommender Algorithms and Mental health*, (Cesifo working papers 2025)

including anxiety and depression. Thus, FoMO clearly has a negative impact on individuals' cognition, mood, and everyday behavior, ultimately intensifying depressive symptoms. Furthermore, studies indicate that FoMO is associated with reduced self-control on social networking platforms, particularly among students. Individuals with high FoMO often experience ego depletion as a result of excessive mobile social media use, leading to reduced self-control and heightened anxiety. The study therefore indicates that social media use produces fear of missing out (FoMO), which in turn contributes to more severe depressive symptoms.[11]

Frequent social media use is associated with increased social anxiety, particularly among young people. This influence is especially strong among individuals born in the twenty-first century, for whom social media has been an integral part of life since birth. People born in this century have not experienced life without social media, which encourages a close connection with it and increases its influence on everyday life. Although social media is commonly believed to help develop social skills among people who find face-to-face communication difficult, excessive use is nevertheless viewed negatively and may adversely affect psychological well-being.[12]

In "Social Media Use and Anxiety in Emerging Adults," the authors examined the relationship between the frequency of social media use and anxiety levels among individuals aged 18–22. Respondents reported the average number of hours per day they devoted to various social media platforms and completed an anxiety questionnaire. Hierarchical regression showed that more time spent on social media was significantly associated with more frequent symptoms of dispositional anxiety. Logistic regression also showed that more frequent daily social media use resulted in higher scores on a measure of anxiety severity, indicating a possible anxiety disorder.[13]

In "No More FOMO: Limiting Social Media Decreases Loneliness and Depression," the authors examined the relationship between the frequency of social media use and mental health problems, including depression. The study sought to identify the changes that occur when individuals temporarily stop using social media or discontinue it for an extended period. A total of 143 students were divided into two groups: one group was restricted to 30 minutes of social

[11] Bin Gao, *why mobile social media-related fear of missing out promotes depressive symptoms? The roles of phubbing and social exclusion,* (BMC Psychology 2023)

[12] O'Day, Emily B., and Richard G. Heimberg. "Social media use, social anxiety, and loneliness: A systematic review." *Computers in Human Behavior Reports* 3 (2021): 100070.
[13] Vannucci, Anna, Kaitlin M. Flannery, and Christine McCauley Ohannessian. "Social media use and anxiety in emerging adults." *Journal of affective disorders* 207 (2017): 163-166.

media use per week, while the other continued using it as usual. The findings showed that depressive symptoms decreased significantly among members of the group whose social media use was restricted, indicating a potentially negative effect of excessive social media use on mental well-being.[14]

In “Impacts of Digital Social Media Detox for Mental Health,” the authors examined the effectiveness of restricting social media use (a detox) for mental health, particularly in relation to depression and stress. The study aimed to assess the impact of deliberately discontinuing social media use. The authors found that, although a detox may have a positive effect on mental health, the issue requires further empirical research and in-depth analysis.[15]

Research Aim, Hypotheses, and Questions

Our study aims to examine the impact of excessive social media use on students’ mental health, particularly in relation to anxiety and depression. With the development of digital technologies, social media has become a platform not only for communication but also for self-expression, access to information, and the formation of social relationships. Despite its diverse opportunities, research suggests that excessive social media use may have serious psychological effects on students. **The study examines whether there is a correlation between the duration of social media use and levels of anxiety or depression. It also considers how negative content intensifies persistent stress and emotional distress. The study will examine the phenomenon of FOMO (Fear of Missing Out), which generates fear of missing important events among social media users and may exacerbate anxiety and depressive states. In addition, it will analyze whether excessive social media use increases social anxiety and how this affects students’ psychological and emotional states. We will also attempt to determine whether temporary or prolonged restrictions on social media use can reduce anxiety and depression.**

Social media undoubtedly plays an important role in students’ lives, serving both as a source of positive connections and as a factor that can harm mental health. Although it is an important means of support and self-expression, excessive use may seriously harm students’ mental

[14] G. HUNT, RACHEL MARX, COURTNEY LIPSON, JORDYN YOUNG, NO MORE FOMO: LIMITING SOCIAL MEDIA DECREASES LONELINESS AND DEPRESSION MELISSA University of Pennsylvania, *Journal of Social and Clinical Psychology*, Vol. 37, No. 10, 2018, pp. 751-768

[15] Ramadhan, Roy N., Derren D. Rampengan, Defin A. Yumnanisha, Sabrina BV Setiono, Kevin C. Tjandra, Melissa V. Ariyanto, Bulat Idrisov, and Maulana A. Empitu. "Impacts of digital social media detox for mental health: A systematic review and meta-analysis." *Narra J* 4, no. 2 (2024): e786.

health. Accordingly, this study will seek to determine whether a correlation exists between students' mental health and excessive use of social networking platforms.[16]

Research Methodology

Sampling

Given the research question, a quantitative research method was employed, specifically the collection of information through a standardized questionnaire. The study's aims and variables made a quantitative approach necessary. The focus was the impact of social media on students' mental health, since students use social media intensively and are at a stage of psychological development. For this reason, the target group comprised students aged 17–24 in Years I–IV at the Free University of Tbilisi and the Agricultural University of Georgia. The total number of students in Georgia is 187,800, which constitutes the study population.[17] Given limitations of time and resources, non-probability sampling was used. Each participant read the questionnaire instructions in advance and took part only after providing informed consent. Confidentiality was fully maintained, and participants could discontinue participation for any reason. Because the term "student" is broad, the questionnaire clearly defined the selection criteria, including the required age range, year of study, and educational institution. Ultimately, 236 students from the Free University of Tbilisi and the Agricultural University of Georgia participated in the study; 72% were female, 27.1% were male, and 0.4% selected "Other" for sex. Of the respondents, 0.4% were aged 17, 7.6% were 18, 19.9% were 19, 34.3% were 20, 28% were 21, 8.1% were 22, 0.8% were 23, and a further 0.8% were over 23. (The complete questionnaire data are presented in the appendix.)

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[16] Douglas T Buzenski, "*The Impact of Social Media on Mental Health in Students*," Capital, accessed March 25, 2025, https://www.capital.edu/admission-aid/college-readiness-resource-hub/mental-health-hub/the-impact-of-social-media-on-mental-health-in-students/.

[17] "Higher Education"- National Statistics Office of Georgia. 2024 https://www.geostat.ge/ka/modules/categories/193/ganatleba

Research Instrument

A questionnaire was developed for the study, comprising five sections that separated the research topics and variables: respondents' personal information, academic activity, social media use, the relationship of social media to anxiety and depression, and social media detox. The questionnaire contained 27 closed-ended questions. Nominal, ordinal, and interval scales were used, enabling respondents to provide their answers as precisely as possible. The questionnaire measured the relationship between two variables, namely social media use and mental health. The study also included other variables, such as sex, age, faculty, year of study, and methods of social media detox. Participants were recruited by email, through which they received the questionnaire, and through the private Facebook group "savele talks," whose members include 7,100 students from the Free University of Tbilisi and the Agricultural University of Georgia, differing in sex, age, and faculty. The survey was administered from May 19 to May 21, 2025, inclusive.

Methods of Data Analysis

At the initial stage of data processing, we used Excel to enter each questionnaire item and the responses received from participants. The information was then coded, allowing us, as the study's authors, to assign a specific numerical value to each response. The next stage involved hypothesis testing and statistical analysis. Particular attention was paid to identifying possible relationships between the variables, which included social media use and mental health. Where a relationship existed, its strength and statistical significance were assessed using indicators such as the **p-value**, **R-squared**, and **Significance F**. We tested the hypotheses using statistical methods including **correlation analysis**, **ANOVA**, **regression analysis**, and the **chi-square test**. These methods enabled a thorough analysis of the data and provided the evidence needed to support or reject the hypotheses.

Study Limitations

Several significant difficulties arose during the research process, limiting the quality of the study and the precision of its interpretation to some extent. One of the main challenges concerned the data collection method: the planned quantitative study was conducted through an online questionnaire, precluding face-to-face communication and close oversight of participants. Consequently, there was a high likelihood that some respondents did not meet the

predetermined selection criteria, limiting the reliability and validity of the data. Some respondents who did not meet the selection criteria nevertheless continued completing the questionnaire, introducing certain inaccuracies into the data. Another important limitation was the sampling method: non-probability sampling was used, which inherently limits the generalizability of the findings to the student population of Georgia. Given the nature of quantitative research, data analysis required statistical competence and effective use of specific analytical tools, which proved a significant obstacle. These factors required the authors to devote additional time and effort to planning and conducting the study. It is therefore essential to consider these limitations when evaluating the results, so that the findings are interpreted within their actual methodological and practical context. Furthermore, a sample of 236 participants does not provide a sufficient basis for drawing general conclusions and generalizing them to students in Georgia, despite the sample's variation in age, sex, and faculty. Technical difficulties were also observed in addition to the methodological challenges.

Analysis of the Research Findings

A total of 236 students completed the questionnaire. Of the respondents, 72% were female and 27.1% were male; 0.4% selected "Other." Within the selected age range of 17–24 years, the largest groups were students aged 20, at 34.3%, and those aged 21, at 28%.

Of the students, 67.8% believe that social media has a negative impact on their mental health, 7.2% believe that social media does not have a negative impact on mental health, 24.2% do not know, and 0.8% declined to answer. (Figure 1)

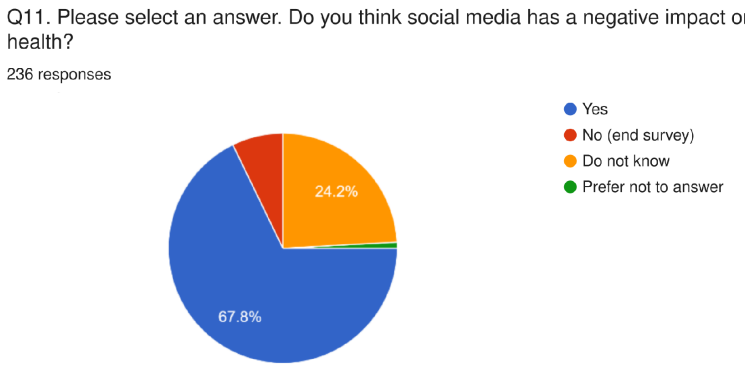


Figure 1

Having established that most students (67.8%) believe that social media negatively affects mental health, we were interested in what they considered the principal reason for this view. Excessive or uncontrolled use emerged as the main reason for the negative impact of social media on students' mental health (37.7%), followed by negative content (15.3%), comparison with others (11.9%), and other reasons. (Figure 2)

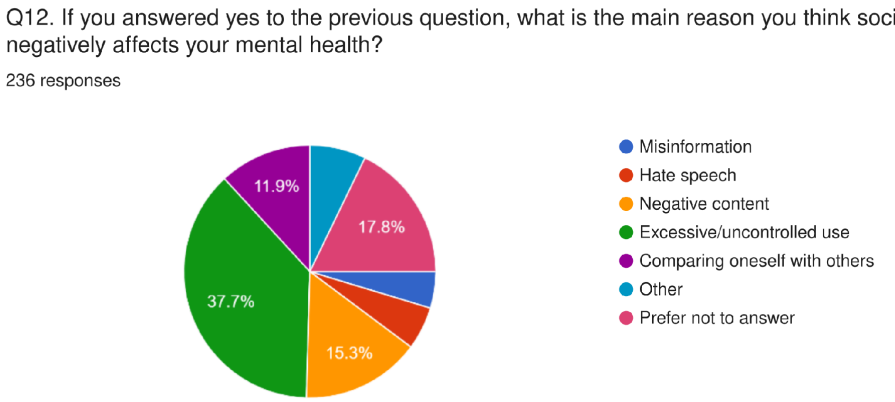


Figure 2

Another interesting finding was that most students had attempted to reduce the time they spent using social media. Of the students, 91.5% answered "Yes" to the question, "Have you ever tried to reduce the time you spend using social media?" (*Figure 3*)

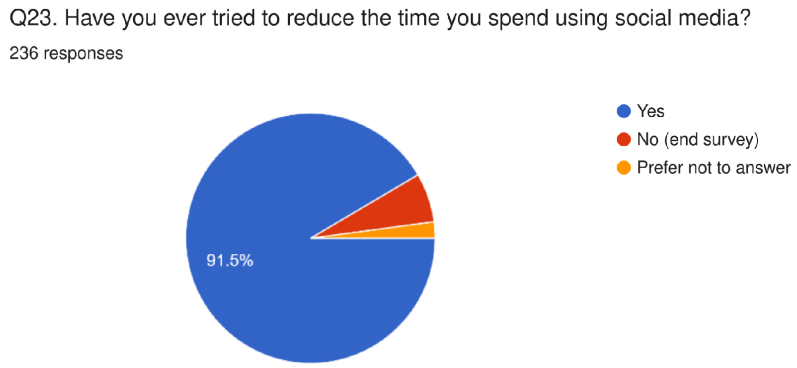


Figure 3

Hypothesis 1: "The more time a student spends on social media, the higher their level of anxiety."

One question in our questionnaire asked how many hours per day students devoted to social media. Their responses showed that 44.1% spent 5–6 hours per day on social media, 43.6%

spent 2–4 hours, 10.2% spent more than 7 hours, and 1.7% spent only 0–1 hour. (See Figure 4.) As we were interested in whether a causal relationship existed between time spent on social media and anxiety levels, we used correlation and regression analyses to test the hypothesis. The correlation, on a scale from −1 to 1, was 0.08, indicating a weak relationship between time spent on social media and anxiety levels. This result suggests that anxiety levels increase as the duration of social media use increases, although the association is very weak. This was the case despite 70% agreeing with the statement, “Time spent on social media increases my anxiety level” (partly agree: 33.5%; completely agree: 15.7%; somewhat agree: 20.8%). Meanwhile, 10.6% completely disagreed and 16.9% partly disagreed. (See Figure 5.) Regression analysis yielded a p-value of **0.169949 > 0.05**, confirming that the hypothesis that greater social media use is associated with higher anxiety levels among students is not supported by these data.

Figure 4

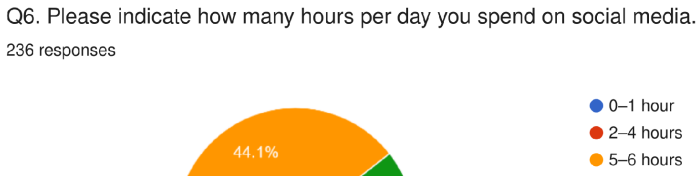
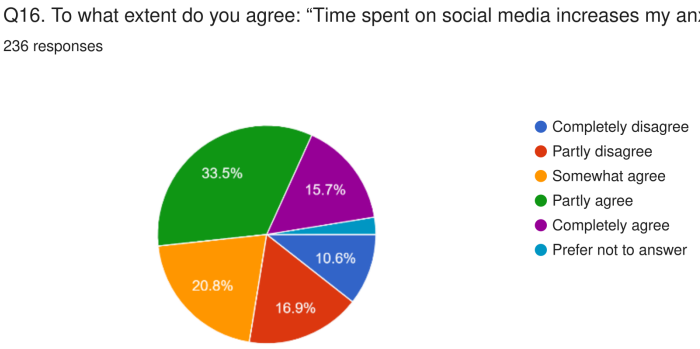


Figure 5



Hypothesis 2: The more time a student spends on social media, the higher their level of depression.

The impact of social media on students’ mental health is not limited to anxiety, so we examined whether the frequency of social media use affects students’ levels of depression. As noted for the preceding hypothesis, 44.1% of respondents spent 5–6 hours per day on social media, 43.6% spent 2–4 hours, 10.2% spent more than 7 hours, and 1.7% spent only 0–1 hour. (See Figure 6.) However, responses to the statement, “Time spent on social media increases my level of depression,” reflected different views on the relationship between increased social media use and increased depression. Of the respondents, 61.9% agreed with this statement (somewhat agree: 25.8%; partly agree: 20.8%; completely agree: 15.3%), while 0.4% declined to answer. (See Figure 7.) Correlation analysis was conducted to test the hypothesis. The

correlation, on a scale from −1 to 1, was **0.024077**, indicating a weak relationship between time spent on social media and levels of depression. This result suggests that students' levels of depression increase slightly as the duration of social media use increases. Regression analysis was conducted to determine whether this relationship was statistically significant and consistent with the hypothesis. The analysis showed that the relationship between time spent on social media and students' levels of depression was not statistically significant, as the overall model p-value was **0.712903 > 0.05**. Accordingly, the hypothesis that greater social media use is associated with higher levels of depression among students is not supported by these data.

Interestingly, most students agree that increased social media use raises both their anxiety and depression levels, yet statistical analysis does not support these hypotheses.

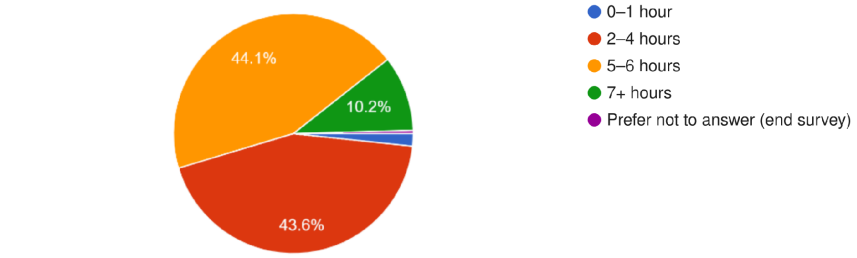


Figure 6

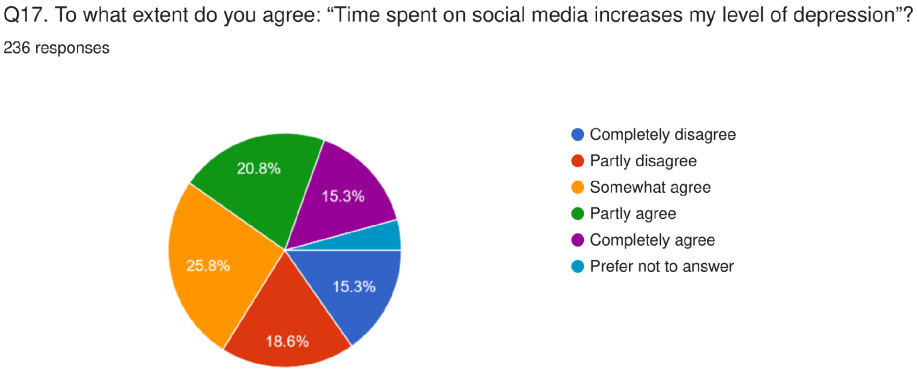


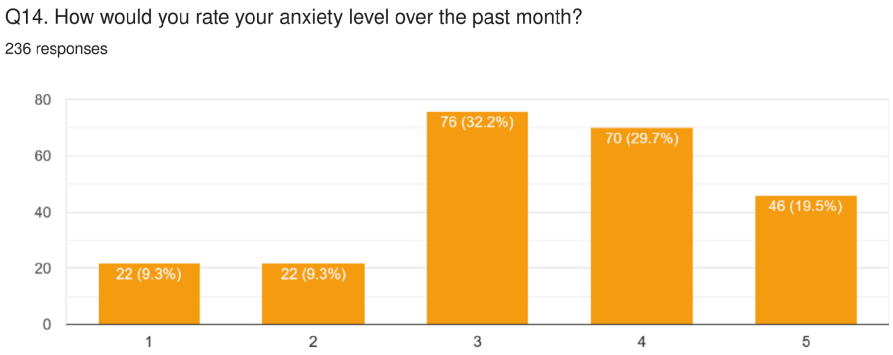
Figure 7

Hypothesis 3: The higher students' level of FOMO when they are not using social media, the higher their level of anxiety.

Fear of missing out on important events (FOMO) is a particularly relevant issue today, especially among students with busy schedules. We therefore examined whether fear of missing important events while not using social media is related to anxiety levels. One questionnaire item asked students to rate their anxiety over the preceding month. The response categories were: 1—very

low, 2—low, 3—moderate, 4—high, and 5—very high. Of the students, 9.3% rated their anxiety over the preceding month as very low, 9.3% as low, 32.2% gave a neutral rating, 29.7% rated it as high, and 19.5% as very high. (See Figure 8.) Students were also asked how strongly they had felt that they were missing important events (FOMO) when not using social media. Of the students, 21.2% had experienced this feeling very mildly, 15.7% had experienced it mildly, and 20.3% gave a neutral response. Of the respondents, 29.7% had experienced a strong fear of missing out while not using social media, and 13.1% had experienced it very strongly. (See Figure 9.) We conducted regression analysis to determine causality and test the hypothesis. The resulting **p-value of 0.0001** < 0.05 indicates a statistically significant relationship between FOMO and anxiety levels. We can therefore conclude that the higher students' perceived FOMO when not using social media, the higher their anxiety levels. This hypothesis is supported by these data.

Figure 8



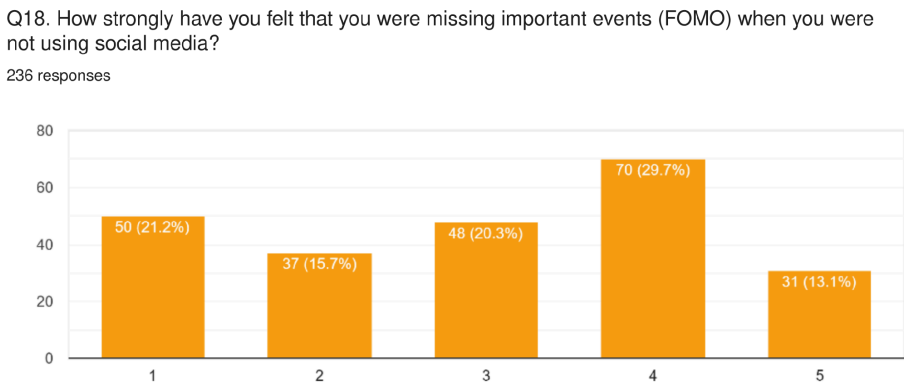
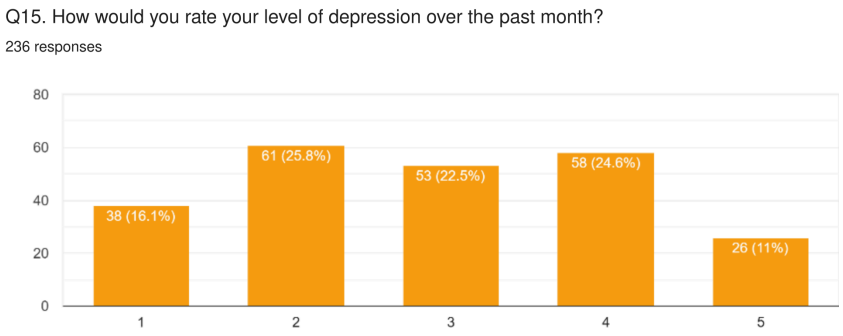


Figure 9

Hypothesis 4: The higher students' level of FOMO when they are not using social media, the higher their level of depression.

Because we were interested in whether fear of missing out on important events (FOMO) affects depression as well as anxiety, respondents were asked to rate their level of depression over the preceding month. Of the respondents, 16.1% rated it as very mild, 25.8% as mild, 22.5% gave a neutral rating, 24.6% rated it as severe, and 11% as very severe. (See Figure 10.) Students were also asked how strongly they had felt that they were missing important events (FOMO) when not using social media. Of the students, 21.2% had experienced this feeling very mildly, 15.7% had experienced it mildly, and 20.3% gave a neutral response. Of the respondents, 29.7% had experienced a strong fear of missing out while not using social media, and 13.1% had experienced it very strongly. (See Figure 11.) We conducted regression analysis to determine causality and test the hypothesis. The resulting **p-value of 0.0249 < 0.05** indicates a statistically significant relationship between FOMO and depression levels. We can therefore conclude that the higher students' perceived FOMO when not using social media, the higher their level of depression. This hypothesis is supported by these data.

Figure 10



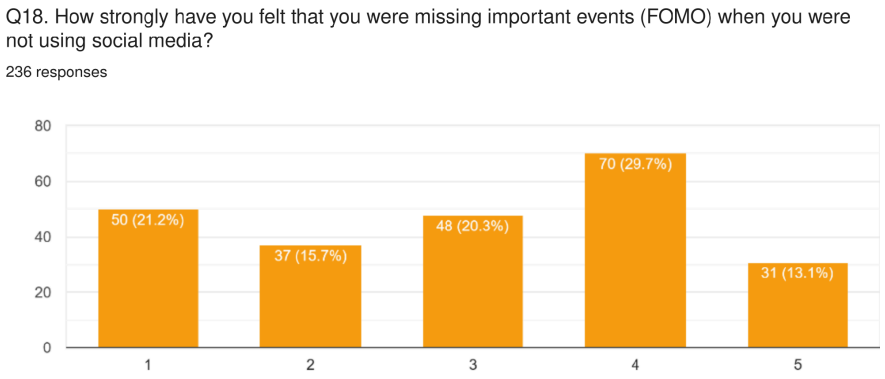


Figure 11

Hypothesis 5: Students who frequently encounter negative content on social media experience emotional distress (irritability, anxiety, sadness, or stress).

The chi-square test was used to examine this hypothesis, allowing the association between two categorical variables to be assessed. In this study, the test assessed whether students' experience of emotional distress (irritability, anxiety, sadness, or stress) in response to content viewed on social media was related to the frequency of viewing negative content during the week. Both variables were initially represented by scales with five or more categories, which made it impossible to apply chi-square analysis directly. For this reason, the variables were recoded into new categorical groups as follows: agreement with the statement that there had been an occasion when content viewed on social media significantly changed the respondent's mood was coded as disagree = 0 and agree = 1. Responses to the question, "How often during the week do you see posts that cause you irritability, anxiety, sadness, or stress?" were coded by frequency of exposure as low (never) = 1, moderate (5–15 times) = 2, and high (15 times or more) = 3. In response to the statement that there had been an occasion when content viewed on social media significantly changed their mood, 4.2% of respondents completely disagreed, 9.3% partly agreed, and 83.9% agreed: 16.5% somewhat agreed, 28% partly agreed, and 39.4% completely agreed. (See Figure 12.) Regarding how often respondents saw posts that

caused irritability, anxiety, sadness, or stress during the week, 26.3% reported seeing such posts 5–10 times, 22.5% reported 10–15 times, 11.9% reported 15–20 times, and 12.7% reported more than 20 times; 1.3% declined to answer. (See Figure 13.) The chi-square test identified a statistically significant association between the frequency of viewing negative content on social media and the experience of emotional distress, **p-value = 9.75958695281539E-15 < 0.05**. This indicates that students who frequently see posts that cause irritability, anxiety, sadness, or stress experience emotional distress more often. Accordingly, the hypothesis that negative content increases emotional distress among students was supported.

Figure 12

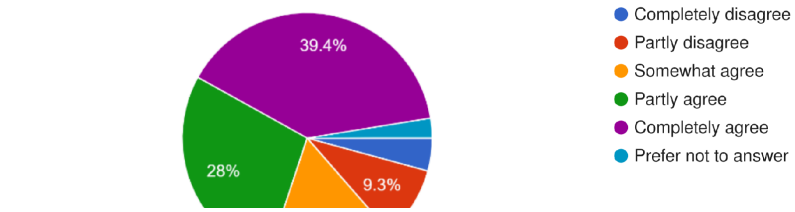
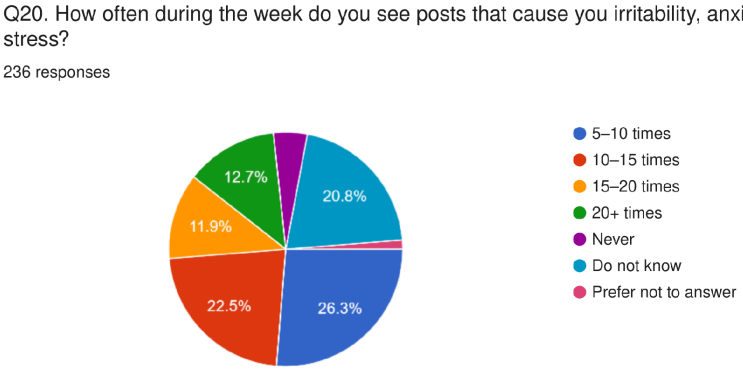


Figure 13



Hypothesis 6: Students who use social media intensively have higher levels of social anxiety than those who use it less frequently.

A one-way ANOVA was used to test the hypothesis, as the aim was to compare mean levels of social anxiety across several groups. To conduct this analysis, respondents were divided into four groups according to the independent variable, intensity of social media use, producing the categorical variable required for ANOVA. The groups were defined as follows: Group 1—students spending 0–1 hour per day on social media; Group 2—2–4 hours; Group 3—5–6 hours; and Group 4—more than 7 hours. The dependent variable was the level of social anxiety, which students rated on a five-point scale. During data processing, social anxiety ratings for

each group were entered in separate columns, allowing mean anxiety levels to be compared across groups. Responses to the question about daily time spent on social media showed that 44.1% of respondents spent 5–6 hours per day, 43.6% spent 2–4 hours, 10.2% spent more than 7 hours, and 1.7% spent only 0–1 hour. (See Figure 14.) In response to the statement, “Social media use increases my social anxiety” (anxiety about communication in real life and fear or anxiety about public communication), 23.7% completely disagreed, 19.9% partly disagreed, 18.6% somewhat agreed, 22.9% partly agreed, and 10.6% completely agreed; 4.2% declined to answer. (See Figure 15.) The one-way ANOVA yielded **p = 0.867 > 0.05**, indicating that mean levels of social anxiety did not differ significantly between groups with different frequencies of social media use. Accordingly, the hypothesis that intensive users have higher social anxiety was not supported. These findings suggest that, in the present data, social anxiety is not significantly associated with the amount of time spent on social media.

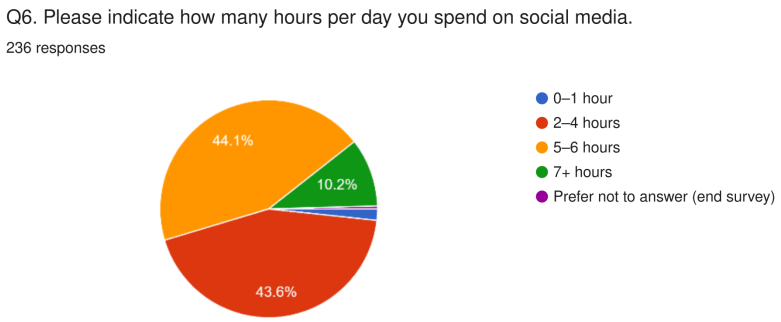


Figure 14

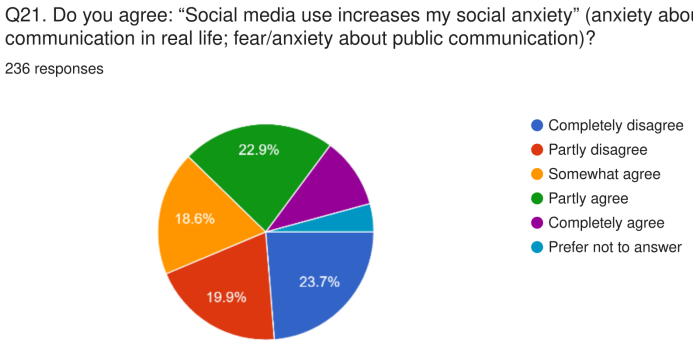


Figure 15

Hypothesis 7: Temporarily restricting social media use reduces students’ anxiety levels.

We were interested in respondents’ views on whether reducing social media use would reduce anxiety levels. One questionnaire item asked students to rate their anxiety over the preceding month. The response categories were: 1—very low, 2—low, 3—moderate, 4—high, and 5—very high. Of the students, 9.3% rated their anxiety over the preceding month as very low, 9.3% as low, 32.2% gave a neutral rating, 29.7% rated it as high, and 19.5% as very high. (See Figure 16.) Respondents were also asked to indicate their agreement with the statement, “Reducing social media use will reduce anxiety levels.” Of the respondents, 2.5% completely disagreed, 8.5% partly disagreed, 27.5% somewhat agreed, 36.4% partly agreed, and 19.5% completely agreed; 5.5% declined to answer. (See Figure 17.) To test the hypothesis and establish a causal relationship between these variables, we conducted regression analysis. The results showed that the variable did not have a statistically significant effect on students’ anxiety levels. The **p-value of 0.6251 > 0.05** indicates that the relationship is attributable to chance and that there is no statistically reliable relationship between the variables. Accordingly, the hypothesis that reducing social media use is associated with reduced anxiety is not supported by these data.

Figure 16

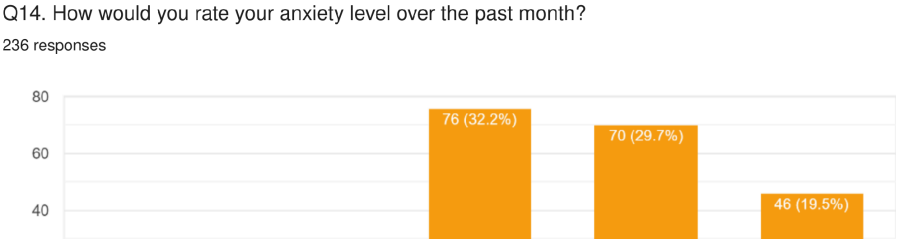
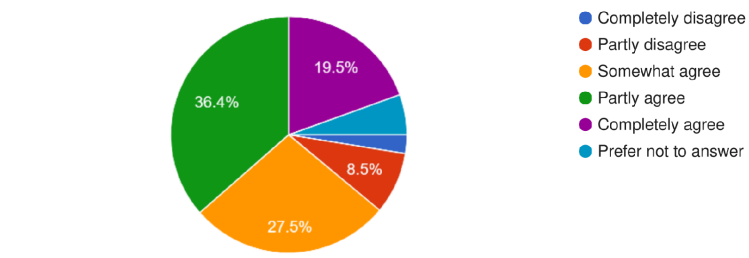


Figure 17

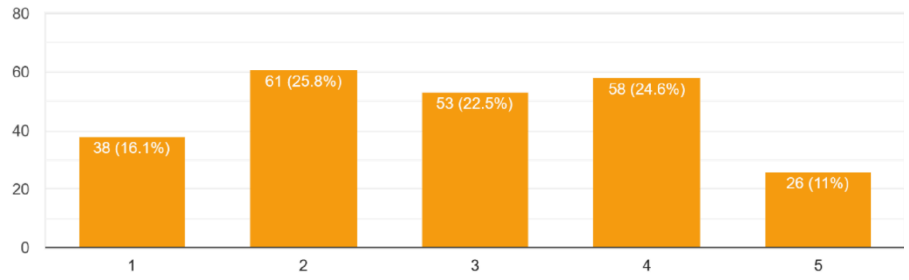


Hypothesis 8: Temporarily restricting social media use reduces students’ levels of depression.

We were interested in whether restricting social media use and undertaking a detox would reduce students’ levels of depression. Accordingly, one questionnaire item asked respondents to rate their level of depression over the preceding month. Of the respondents, 16.1% rated it as very mild, 25.8% as mild, 22.5% gave a neutral rating, 24.6% rated it as severe, and 11% as very severe. (See *Figure 18*.) Respondents were also asked to indicate their agreement with the statement, “Reducing social media use will reduce levels of depression.” Of the respondents, 3.4% completely disagreed, 12.3% partly disagreed, 27.1% somewhat agreed, 34.3% partly agreed, and 17.3% completely agreed; 5.1% declined to answer. (See Figure 19.) To test the hypothesis and establish a causal relationship, we conducted regression analysis. The results showed that the variable did not have a statistically significant effect on students’ levels of depression. The **p-value of 0.8988 > 0.05** indicates that the relationship is attributable to chance and that there is no statistically reliable relationship between the variables. Accordingly, the hypothesis that reducing social media use is associated with lower levels of depression is not supported by these data. *Figure 18*

Q15. How would you rate your level of depression over the past month?

236 responses



Q26. To what extent do you agree: "Reducing social media use will reduce levels of depression"?

236 responses

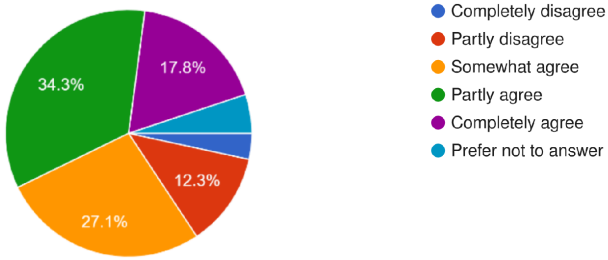


Figure 19

Conclusion

Contrary to expectations, the study did not establish a relationship between the time students spent on social media and their levels of anxiety and depression. This finding may indicate that duration of use alone does not determine psychological health. In other words, time and use do not automatically affect mental health, raising the question of which component is decisive in linking social media to mental health difficulties. The part of the study examining the influence of FOMO, or fear of missing out on important events, provides an important contribution to answering this question. Students who strongly experience the feeling of being without social media showed higher levels of anxiety and depressive symptoms. This reflects emotional dependence on the digital world, which appears far more important than platform use frequency alone. The nature of the content proved particularly important. Students who frequently

encounter negative posts experience greater emotional distress, irritability, anxiety, sadness, or stress. This emphasizes that the content-related aspect of social media is directly linked to emotional well-being. A digital environment filled with tension, excessive negative information, and social pressure can be mentally exhausting and harmful to students. The conclusions regarding social anxiety are somewhat different. Contrary to expectations, no clear difference in social anxiety levels emerged between intensive social media users and those who were less active. This may indicate that social anxiety depends more on individual personality characteristics than on time spent on social media. The study also failed to demonstrate a clear relationship between students' mental health and temporarily restricting social media use. Even among students who agreed that spending less time on social media could positively affect mental health, no actual differences in anxiety or depression levels were identified. The impact of social media on students' mental health cannot be explained solely by the frequency or duration of platform use. The psychosocial processes occurring in the digital environment are considerably deeper and more complex.

Recommendations

For those planning future research on the impact of social media on students' mental health, several methodological and substantive recommendations should be considered to improve research accuracy and the generalizability of the results. First, probability sampling is recommended to ensure that participants are representative and to allow the findings to be generalized to the entire student population. Random sampling reduces bias and increases research reliability in relation to two principal criteria of quantitative research: precision and generalizability. In addition, a sociodemographically diverse sample is advisable, including respondents from different social, gender, age, and cultural groups. The development of research instruments is another important recommendation. Clinically validated and substantively comprehensive scales should be used to capture more adequately the complexity of variables such as anxiety, depression, and FOMO. Furthermore, quantitative analysis alone is not always sufficient to understand the complexity of social media's effects. Mixed-methods approaches are therefore recommended, combining quantitative methods with qualitative methods such as interviews, focus groups, or open-ended questions to strengthen the depth of interpretation and contextual understanding. Implementing these recommendations would enable future studies to advance the investigation of this topic, contributing substantially both to academic discourse and to effective policies and preventive practices for protecting students' mental health.

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Appendix

Figure 1

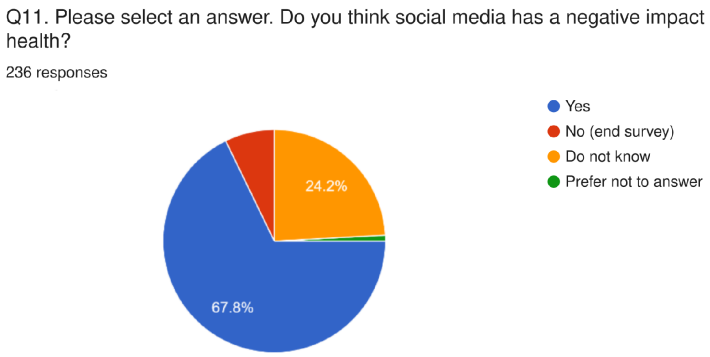


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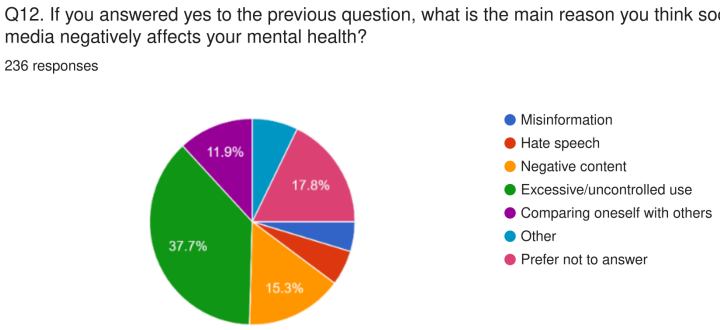


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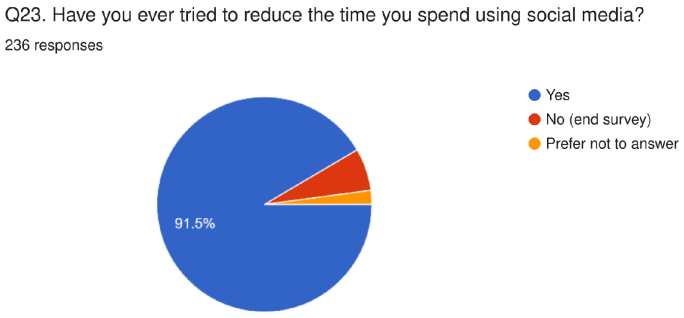


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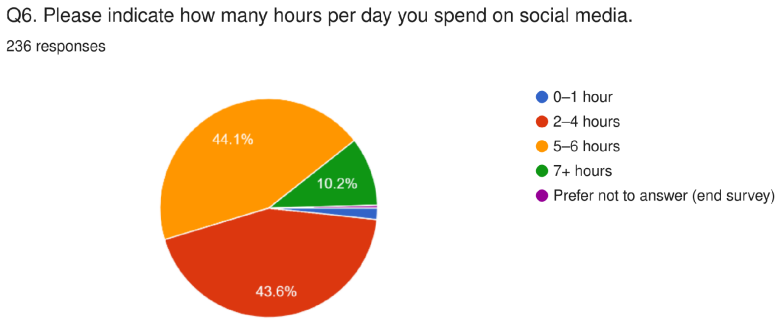


Figure 5

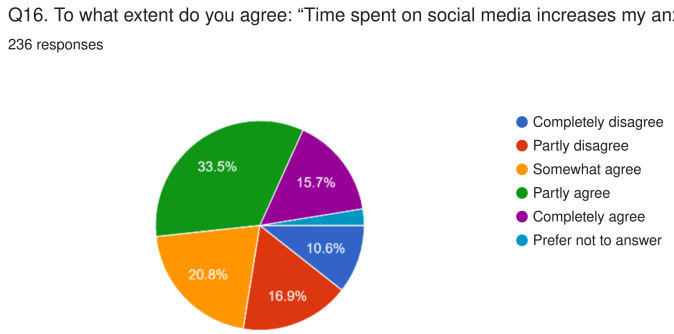


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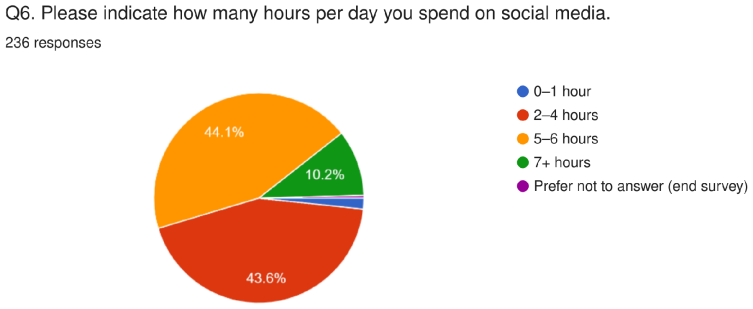


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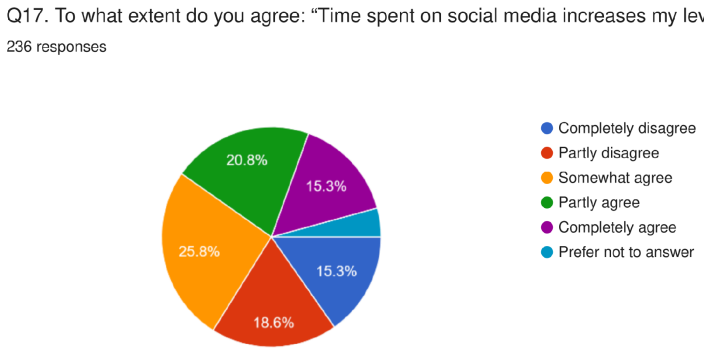


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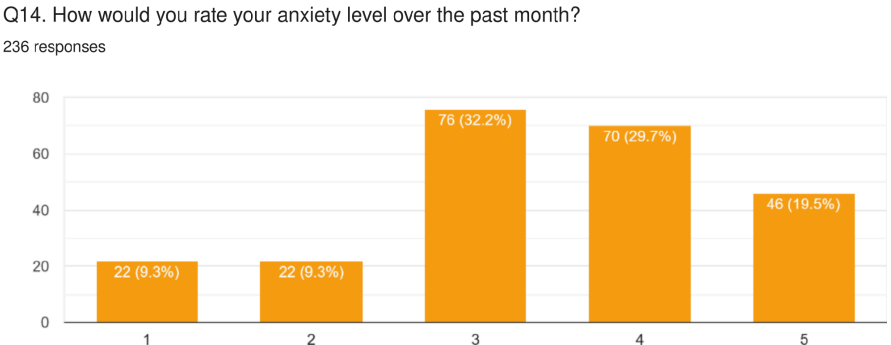


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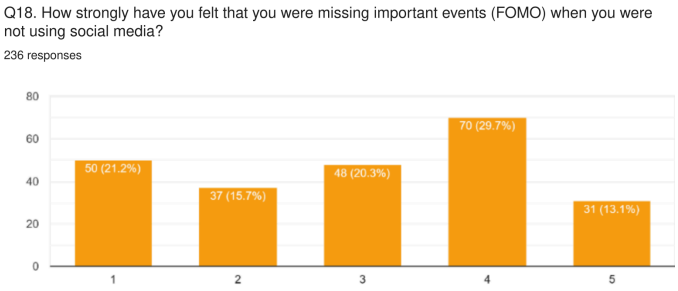


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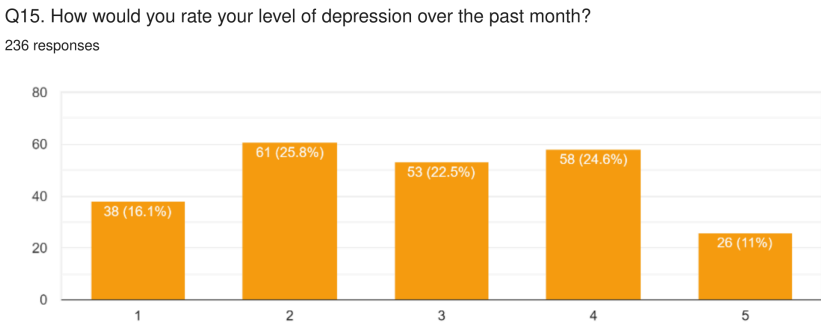


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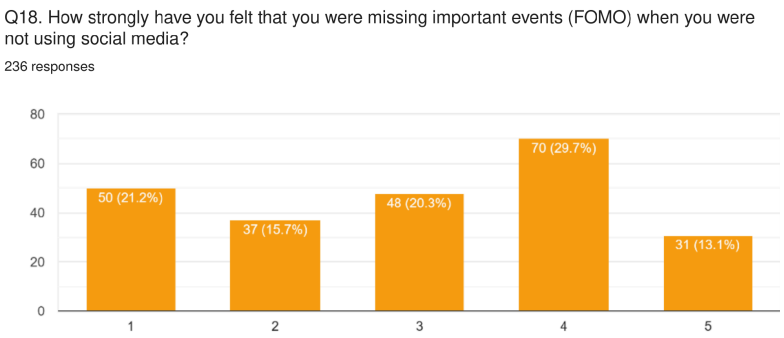


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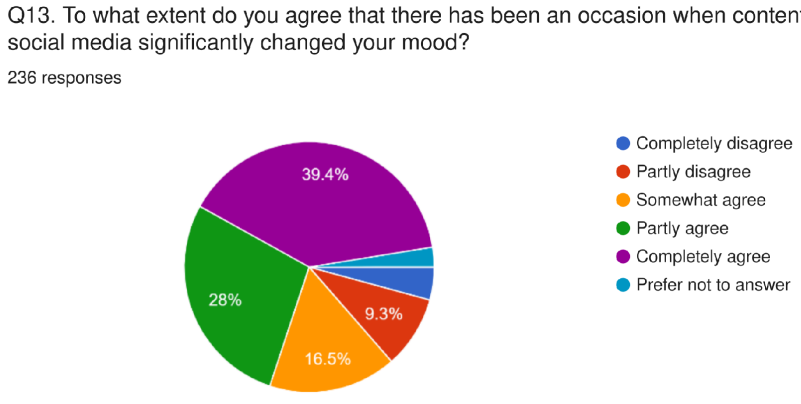


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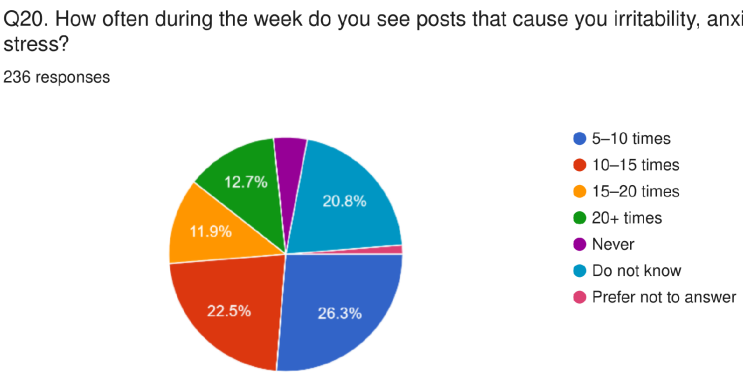


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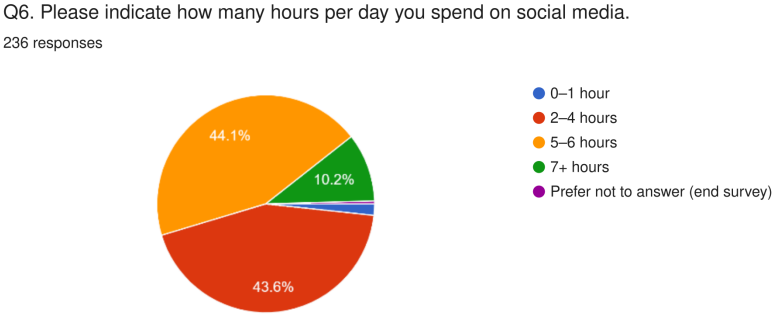


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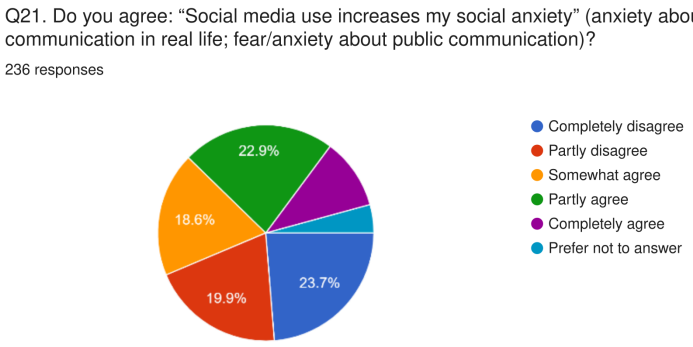


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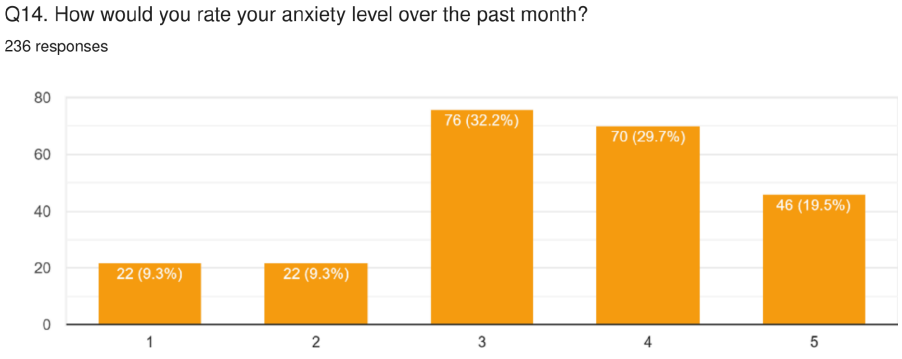


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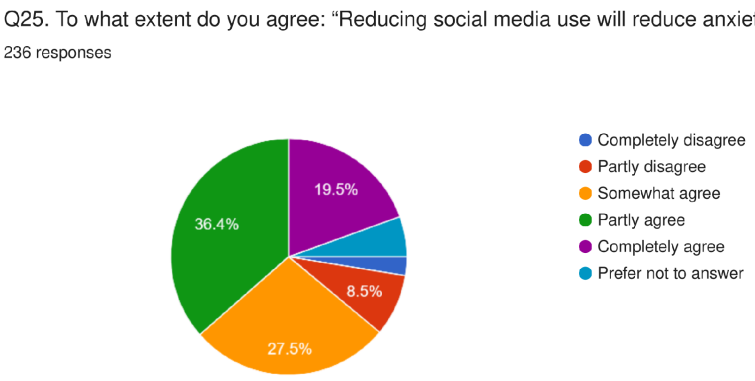


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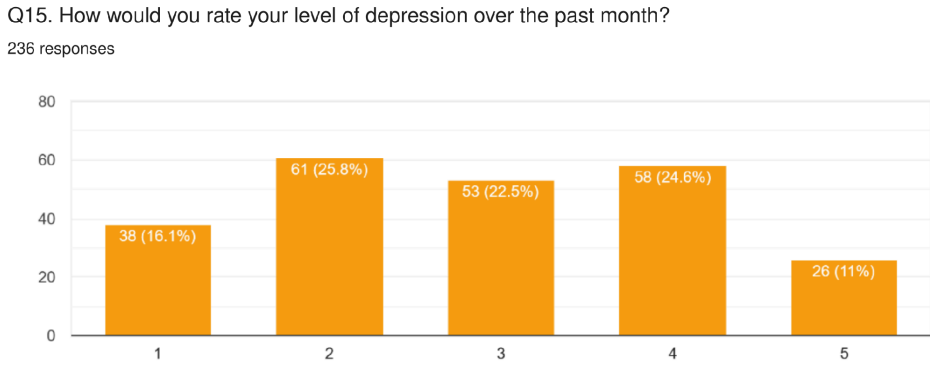
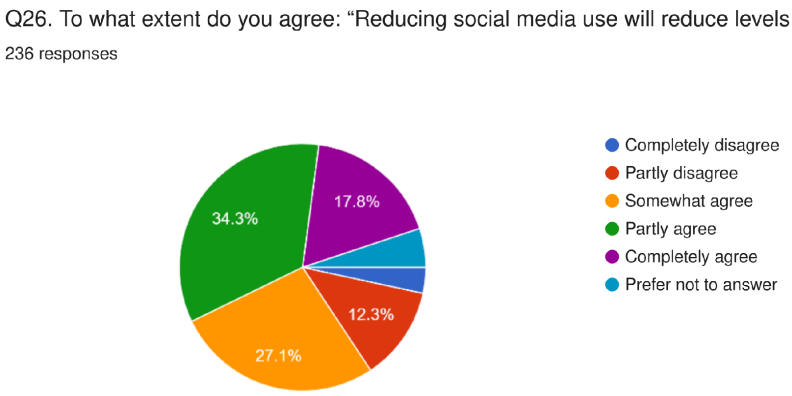
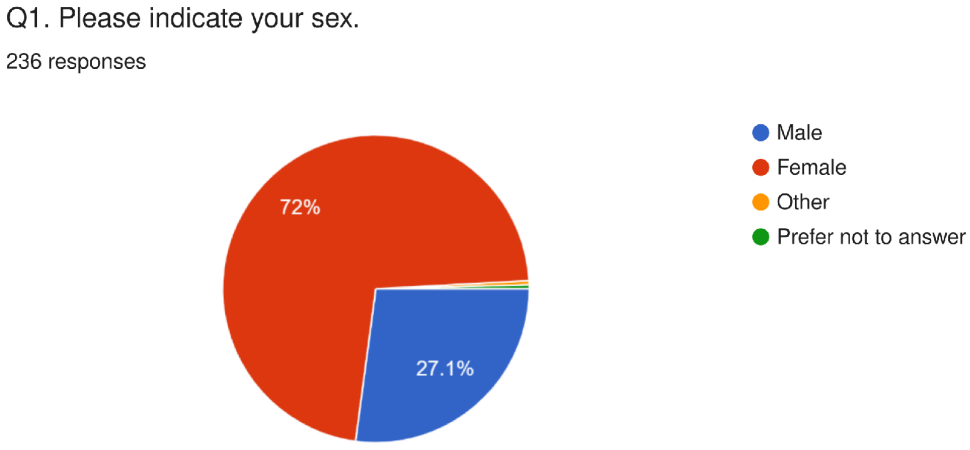


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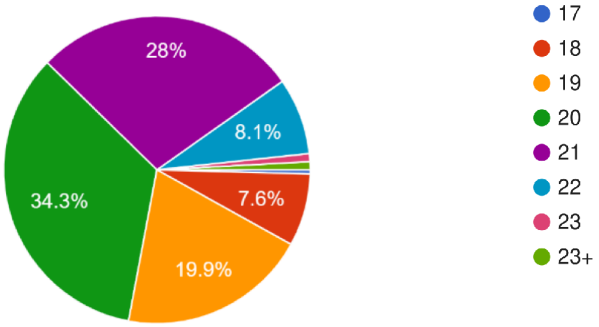


Figures Not Used in the Analysis of the Research Findings



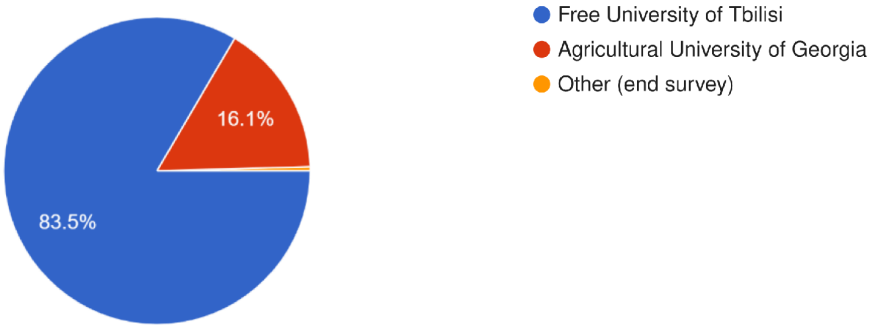
Q2. Please indicate your age.

236 responses



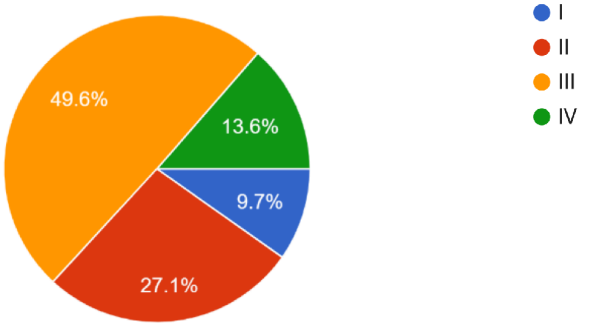
Q3. Please indicate your educational institution.

236 responses



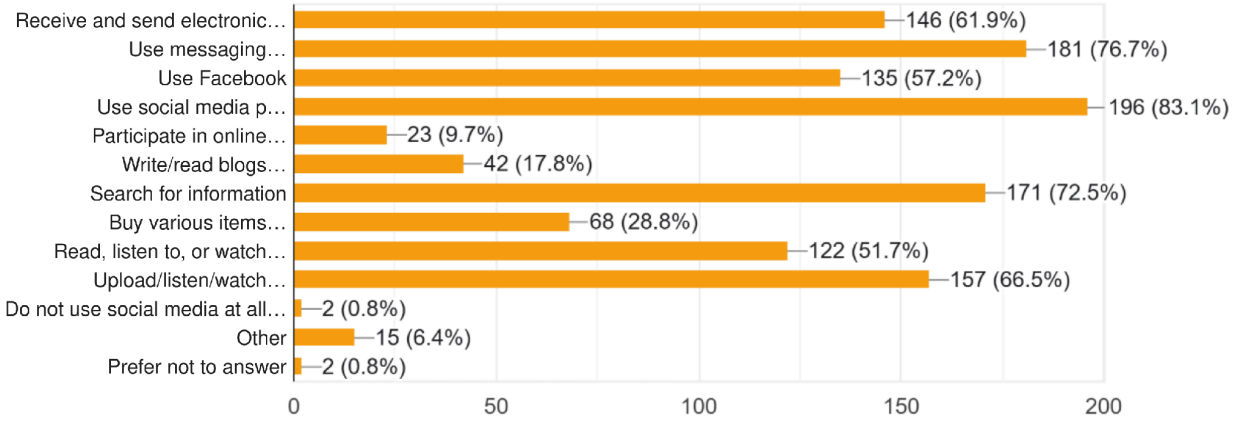
Q5. Please indicate your year of study.

236 responses



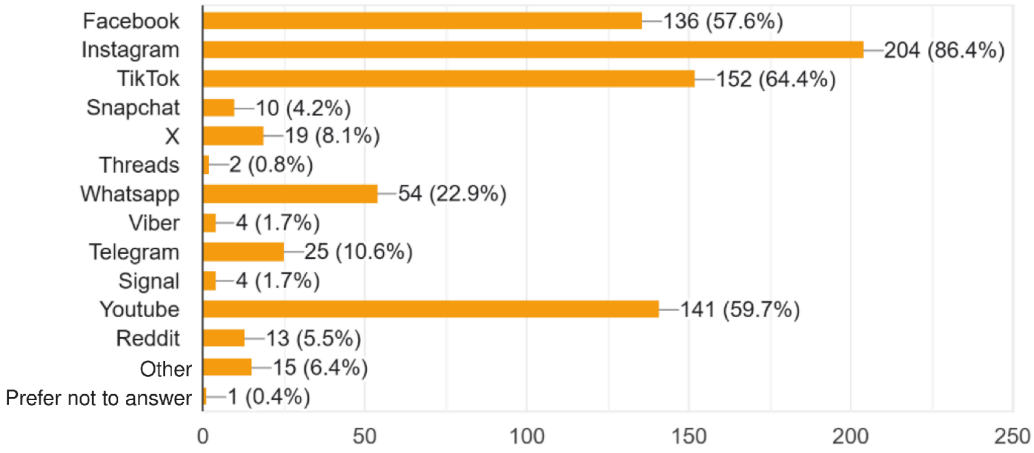
Q7. Please indicate your main purposes for using social media. Select multiple answers.

236 responses



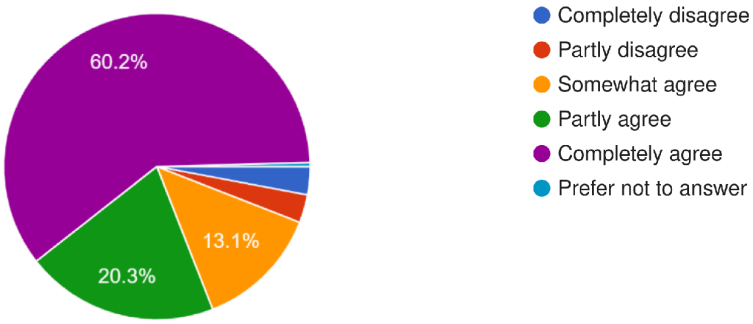
Q8. Please indicate which platforms you use most frequently. Select multiple answers.

236 responses



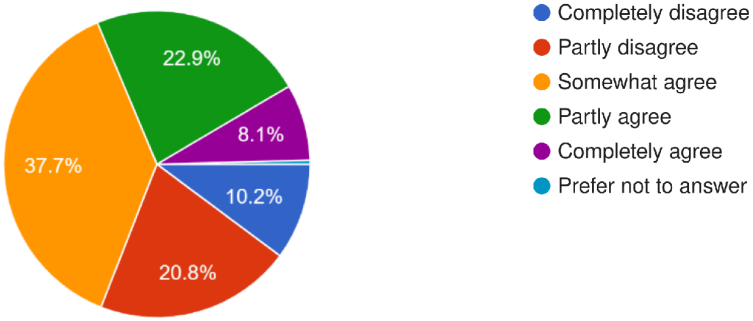
Q9. Please indicate how much you agree: “I spend more time on social media than necessary.”

236 responses



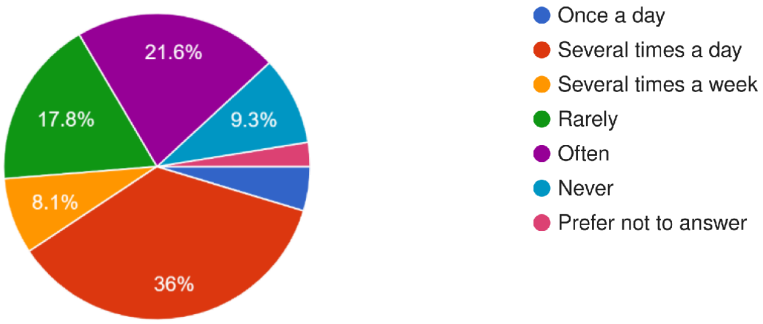
Q10. Please indicate how much you agree: “I can control how frequently I use social media.”

236 responses



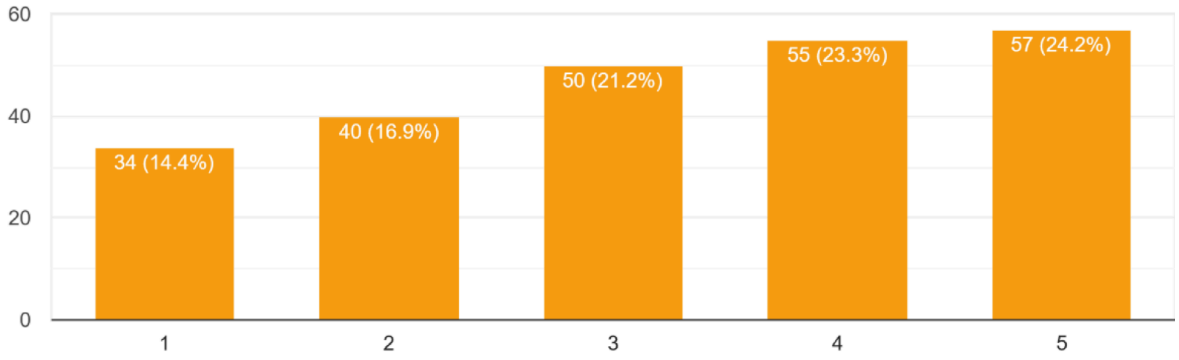
Q19. How often do you check social networking platforms because of FOMO ("fear of missing out")?

236 responses



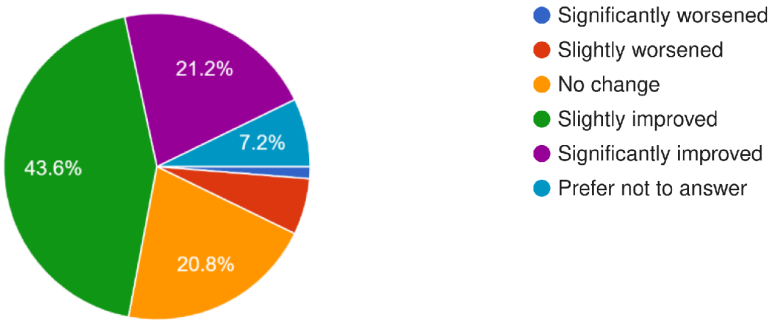
Q22. How strongly do you feel the need to restrict your social media use to improve your mental health?

236 responses



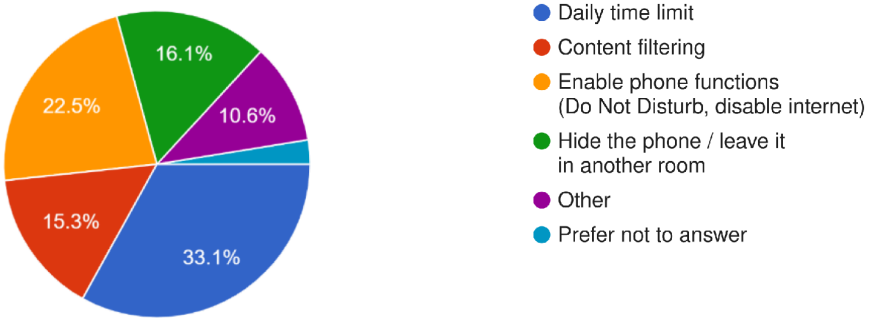
Q24. How did reducing your social media use affect your emotional state?

236 responses



Q27. Which strategy do you think would be most suitable for regulating your social media use?

236 responses



Questionnaire

https://docs.google.com/forms/d/e/1FAIpQLSe5MeTMESOn3EkSktO6XifOnJinU5m9iXUwwt28ttiGLRUFqg/viewform?usp=dialog

See the separate Excel file for the datasets.